Fiscal Year:	FY 2020	**Task Last Updated:**	FY 12/09/2019
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Project Title:	Stationary and Swing-Like Pool Boiling for Microgravity Applications for Cooling of High- Power Microelectronics		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	60607	**Congressional District:**	7
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2009 ESA AO Biology on Sounding Rockets and Physical Sciences on All Platforms
Start Date:	03/01/2017	**End Date:**	02/28/2021
No. of Post Docs:	2	**No. of PhD Degrees:**	4
No. of PhD Candidates:	6	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX17AF33G		
Performance Goal No.:			
Performance Goal Text:			
	NOTE: Continuation of "Spray Cooling in Flooding Regime: Pool Boiling on Nano-Textured Surfaces," grant NNX13AQ77G, Principal Investigator Alexander Yarin, PhD. The European Space Agency (ESA) recently initiated “Thermal Platform 1” research project, a ramification of DOLFIN, aiming research projects relevant for the work of the International Space Station (ISS) and for the further Space exploration. This document foresees that: "In the area of space applications and with the increasing size of space platforms or future exploration missions, large amounts of heat need to be dissipated by large space radiators. Efficient thermal control in space and the reduction of moving mechanical elements become of crucial importance and two-phase closed-loop systems can meet these requirements." The present project supported by NASA will address some of these issues and is planned to become an integral part of the ESA activities. The University of Illinois at Chicago (UIC) group		

Task Description:	supported by NASA in the framework of this project actively collaborated with the ESA supported group at the Technical University Darmstadt (Germany), where a group of 3 Master students, 1 PhD student, and 1 bachelor student were involved. Drop and spray cooling in such situations could offer an attractive alternative, and our previous work, supported by NASA as a part of participation in DOLFIN, was directed on the enhancement of drop cooling using nano-textured surfaces. In spray cooling with surface flooding, heat transfer occurs in the regime of pool boiling. The regimes of pool boiling can be divided on the boiling curve as free convection boiling, nucleate boiling, transition boiling, and film boiling. Among them nucleate boiling is considered to be the most desirable heat transfer regime, since it is associated with high heat fluxes at surfaces with low superheat temperature, which is attractive for such practical applications as cooling of electronics, boilers, nuclear and chemical reactors, and refrigeration systems. However, there exists a critical value of heat flux called critical heat flux (CHF) at which nucleate boiling transforms into film boiling. The latter involves heat transfer through coolant vapor rather than through liquid coolant itself. There were multiple efforts to increase CHF and boiling heat transfer coefficient (BHT), which is by definition the heat flux divided by the corresponding superheat temperature. In particular, boiling on roughened surfaces was found to increase BHT coefficient, which stems from an increased area density of nucleation sites. Microporous coatings to enhance pool boiling were also studied. In our preliminary work on pool boiling on nano-textured surfaces under full gravity conditions, in the ground experiments the nano-textured surfaces used were copper platelets covered with copper-plated electrospun nanofibers. In addition, for comparison, pool boiling on the corresponding bare copper surfaces was experimentally studied. The results revealed that the heat flux and heat transfer coefficient in boiling on the nano-textured surfaces were about 3-8 times higher than those on the bare copper surfaces. This stems from the fact that nano-textured surfaces promote bubble growth by increasing the average temperature of fluid surrounding growing bubbles, as our experimental and theoretical results show. Hence, nano-textured surfaces manufactured from copper-plated nanofibers facilitate bubble growth rate and, thus, increase bubble detachment frequency. The critical superheat temperature of nano-textured surfaces was about 10 degrees C lower than that on smooth surfaces. In addition, the heat flux on the nano-textured surfaces in transition boiling was significantly higher than on the bare copper ones, since the presence of nanofibers prevented bubble merging and delayed formation of vapor film. [ED. NOTE (12/2017): initial project from this work was " 'DOLFIN' DROP IMPACT ON POROUS NANO-TEXTURED SURFACES," (grant NNX10AR99G).]
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The proposed method incorporates two main novel elements: nanofibers at the heater surface and self-propelled swing-like motion of the heater. These nanofibers provide additional nucleation sites, resulting in much more frequent bubble nucleation and departure, and thus a higher overall vapor recoil force propelling the heater motion. In turn, the self-propelled heater motion facilitates shedding of vapor layer and thus, delaying the critical heat flux. The resulting intensification of the heat removal rate and delay of the critical heat flux are characteristic of cooling by means of pool boiling not only in microgravity but also on Earth, under normal gravity. Accordingly, the proposed approach can be broadly applicable to multiple cases of cooling of high-power microelectronics and other devices not only in space but also in a wide range of ground application, which determines the broader impact of the proposed approach.
Task Progress:	[Editor's note (12/9/2019): compiled from PI's FY2020 annual progress received December 2019] Pool boiling for of effective cooling of high-power microelectronics is a highly desirable method in ground and space applications. However, it is strongly limited by the inception of the critical heat flux (CHF) when a practically intact vapor layer covers the heater surface and prevents further heat removal. In microgravity, the CHF sets in much earlier because there is no buoyancy force which facilitates vapor removal from the heater surface. Here, we explore the method which allows vapor removal from the heater surface on ground and in microgravity. The method uses a part of the removed heat to sustain a suspended heater motion. Accordingly, without any additional motors (and thus, additional dissipation), heater escapes from the adjacent vapor layer, which prevents/delays the incipience of the CHF. In particular, a single submerged heater with one side of it covered by supersonically-blown 50 nm nanofibers will facilitate asymmetric bubble nucleation and detachment on that side. That creates a thrust force (a bubble recoil force) acting on the heater. The heater is suspended and thus, begins moving. As a result the vapor layer shedding begins and delays reaching the critical heat flux (CHF). Sandwiched suspended heaters can be powered alternatively resulting in a swing-like motion. The preliminary terrestrial experiments strongly support this mechanism. This approach is expected to dramatically facilitate pool-boiling cooling of high-power microelectronic chips under microgravity conditions. Not only the vapor layer is effectively shed off the heater surface in this approach irrespective of the gravity force, but also the swing-like motion of a suspended chip or device (a heater) mixes the coolant and thus additionally facilitates the heat removal and dissipation. It should be emphasized that this mechanism employs a part of thermal energy removed from the heater surface to produce mechanical work before the ultimate dissipation, and does not require any additional power-consuming devices for mechanical stirring or electrical actuation. Moreover, vapor recoil thrust does not depend on the presence of gravity and the buoyancy force, and thus should be operational in microgravity.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Articles in Peer-reviewed Journals	Sankaran A, Yarin AL. "Evaporation-driven thermocapillary Marangoni convection in liquid layers of different depths." International Journal of Heat and Mass Transfer. 2018 Jul;122:504-14. https://doi.org/10.1016/j.ijheatmasstransfer.2018.01.136 , Jul-2018
Articles in Peer-reviewed Journals	Sankaran A, Zhang W, Yarin A. "Pool boiling in deep and shallow vessels and the effect of surface nano-texture and self-rewetting." International Journal of Heat and Mass Transfer. 2018 Dec;127:857-66. https://doi.org/10.1016/j.ijheatmasstransfer.2018.08.046 , Dec-2018